

Lab 24 — Locally Weighted Regression (LWR) from Scratch

Aim

To implement Locally Weighted Regression (LWR) to fit a non-linear curve using a Gaussian kernel-weighted local linear model.

Algorithm / Procedure

1. Generate noisy sine-wave data.
2. For each query point, calculate Gaussian weights based on the distance from the training points.
3. Construct the weighted design matrix.
4. Calculate the local regression parameters using weighted least squares.
5. Predict the output for each query point.
6. Plot the original data and the LWR fitted curve.
7. Save the plot as `lwr_fit.png`.

Program

```
import numpy as np

import matplotlib.pyplot as plt

def lwr(x_query,X,y,tau=0.5):

    m=X.shape[0]

    W=np.exp(-((X-x_query)**2)/(2*tau**2))

    Xb=np.c_[np.ones(m),X]

    W=np.diag(W)

    theta=np.linalg.pinv(Xb.T@W@Xb)@Xb.T@W@y

    return np.array([1,x_query])@theta

np.random.seed(1)

X=np.linspace(0,10,100)
```

```
y=np.sin(X)+np.random.normal(0,0.1,100)

y_pred=np.array([lwr(x,X,y,tau=0.5) for x in X])

plt.scatter(X,y,s=10,label='Data')

plt.plot(X,y_pred,label='LWR Fit')

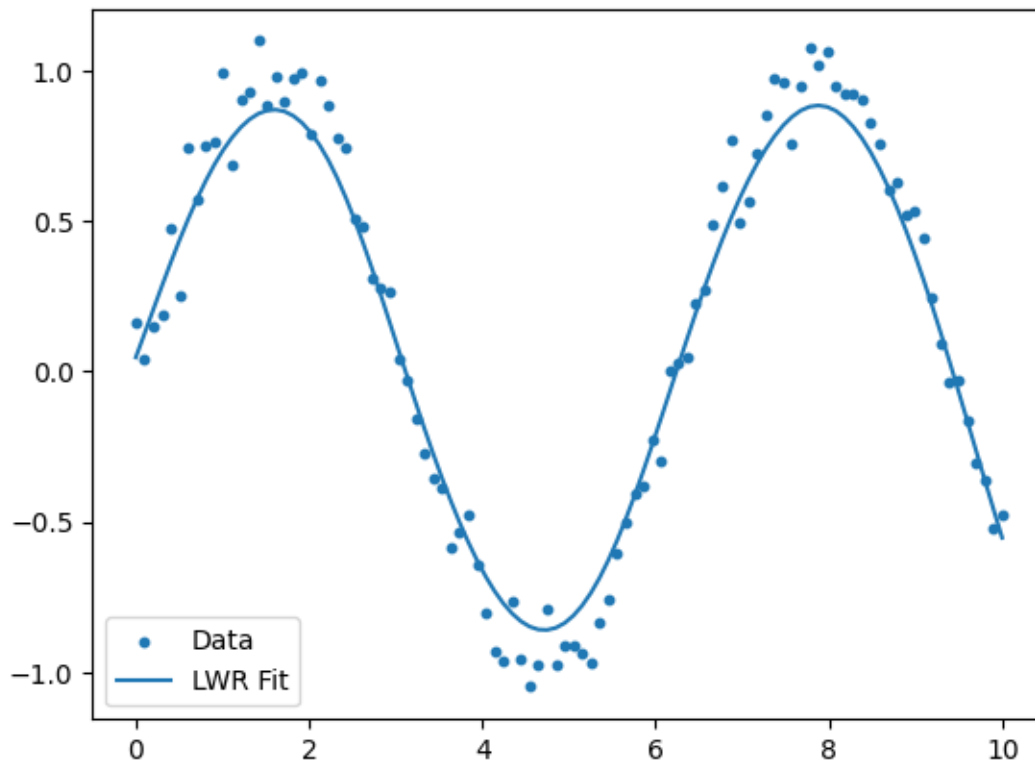
plt.legend()

plt.savefig('lwr_fit.png')

print("Sample predictions:",np.round(y_pred[:5],3))
```

Output

Sample predictions: [0.047 0.127 0.207 0.286 0.364]



Result

Locally Weighted Regression was successfully implemented from scratch using a Gaussian kernel. The model produced a smooth non-linear curve that closely followed the noisy sine-wave data.